

HAT-P-32b

R-1338

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_211110 · created 2026-07-31T21:18:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459526.7113 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.051
Transit depth (Rp/Rs)^2	3.45 +/- 1.9 [%]
Orbital Inclination (inc)	89.54 +/- 1.71
Airmass coefficient 1 (a1)	20.52 +/- 2.26
Airmass coefficient 2 (a2)	-2.99 +/- 0.23
Scatter in the residuals of the lightcurve fit is	12.01 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.0
Transit Duration (day)	0.0598 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.051 vs 0.1489 (deviation 0.48σ)
Duration fit vs published	0.0598 d vs 0.1304 d (deviation 9.62σ)
Mid-time O-C	-3103.3 min
Depth SNR	2.4
Residual scatter	12.01 %

Light curve

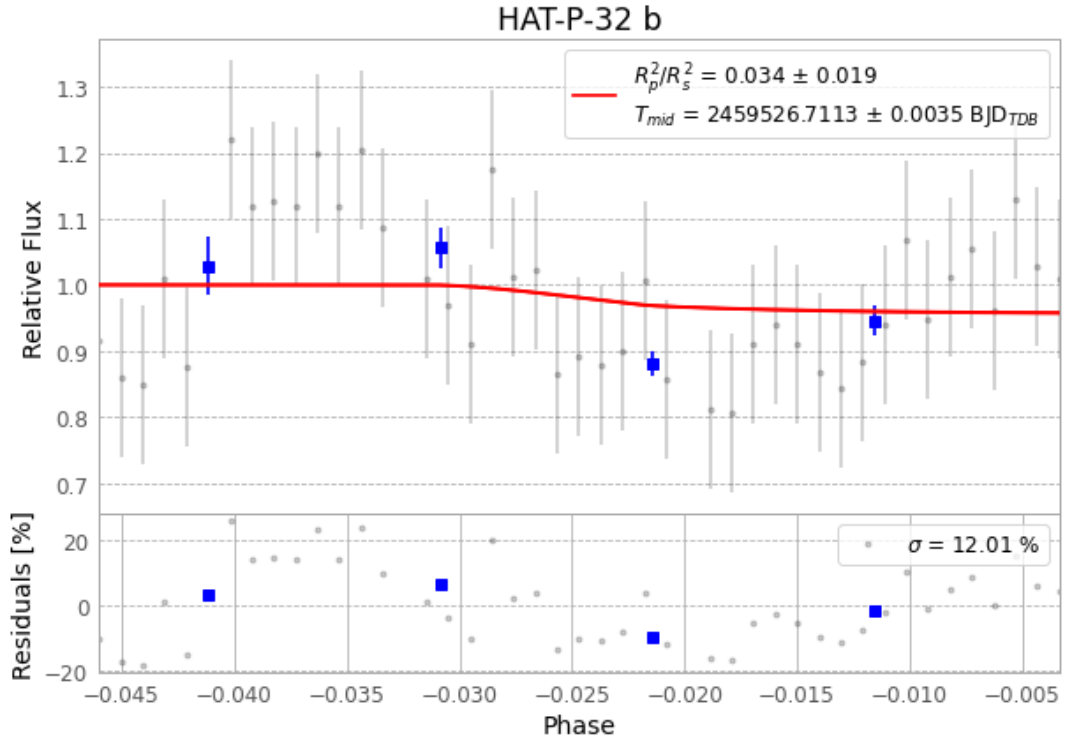


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-11-09.png (SHA-256 d1add397bd2ce82b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [145, 60], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 308e1ccd59ae7f6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

45 FITS frames (30 MB), HATP-32211110060912.FITS → HATP-32211110082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-211110.log (16d26db903b6...), FinalLightCurve_HAT-P-32 b_2021-11-09.png (d1add397bd2c...), FinalLightCurve_HAT-P-32 b_2021-11-09.csv (1683805b898d...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:18Z.